

Misbeliefs and Education Decisions by Gender

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Big Picture

This paper motivated by 3 facts

- ▶ Women **earn more** bachelor's degrees than men in all 38 OECD countries. In the U.S., the gap is 47% vs. 37%, and men **drop out more**.
- ▶ Males **overpredict** academic performance more than females, especially low-performing males (Owen, 2023; Lundeberg et al., 1994).
- ▶ If beliefs drive enrollment decisions, **gendered misbeliefs** could generate over-enrollment, dropout, and misallocation of talent.
- ▶ **Key Question:** How much of the education gender gap can be explained by belief differences between men and women?
- ▶ Misbeliefs could distort education decisions, leading to talent and resource misallocation, wasted tuition, dropout debt, lower graduate quality and welfare losses.

in this paper I argue that some part of GEG originates even before students set foot on campus in what they think about themselves

What I Do

to progress with this question

- ▶ Document **subjective graduation beliefs** by gender and compare with **actual outcomes**.
- ▶ Provide empirical evidence that **how beliefs** could effect education decisions.
- ▶ Build a **GE education choice model** with subjective beliefs.
- ▶ Show how misbeliefs generate over-enrollment and dropout by gender.
- ▶ Counterfactual: belief correction.

What I Find

my results shows that

- ▶ Females and males are both **overconfident** about graduation.
- ▶ Females report **higher beliefs** and have **higher graduation rates**.
- ▶ **Males are less accurate**: larger belief–outcome gap after controlling for skill.
- ▶ Men's decisions are **more belief-elastic**, and their beliefs are **less informative** — so they act more on **unreliable signals**.
- ▶ Misbeliefs could contribute to **male dropout** and **female dominance among graduates**.

Related Literature *This paper is related to*

► **Gender gap in educational attainment.**

- Feng, Ren, and Rendall (2023) *but they don't have belief*

► **Overconfidence differs by gender.**

- Lundeberg, Fox, and Punčochař (1994)
- Owen (2023)
- Barber and Odean (2001)
- Cortés, Pan, Pilossoph, Reuben, and Zafar (2023)

► **Beliefs shape education outcome.**

- Stinebrickner and Stinebrickner (2012, 2014)
- Arcidiacono, Hotz, and Kang (2012)
- Wiswall and Zafar (2015)
- Moschini, Raveendranathan, and Xu (2024)

This paper: Gendered misbeliefs → education decisions.

Empirical Evidence

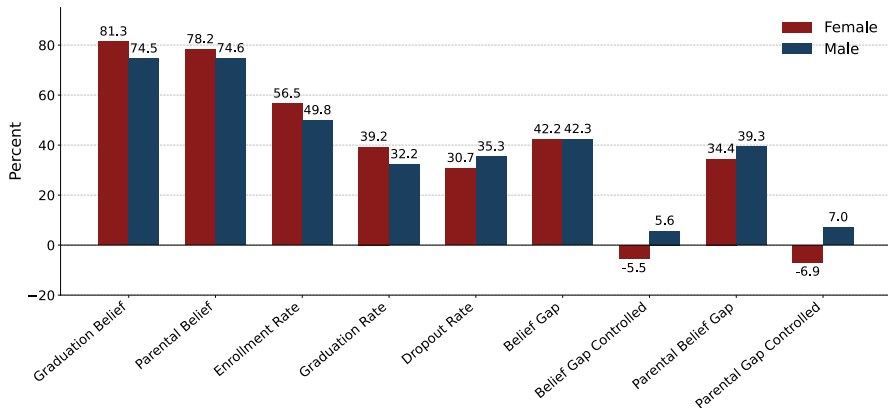
Data: NLSY97

for data I used

- ▶ **Source:** National Longitudinal Survey of Youth 1997 (NLSY97). Nationally representative, ~9,000 individuals born 1980–1984.
- ▶ **Key variables:**
 - student belief: self-reported % chance of completing a 4-year degree by age 30, elicited in 1997.
 - skill: ASVAB composite score percentile.(math, verbal, science & technical, mechanical & spatial)
 - effort: self-reported effort in school (scale 1–10).
 - risk tolerance: self-reported willingness to take risks (scale 1–10).
 - parental belief: parent-reported % chance child completes a 4-year degree by age 30.
 - enrollment status: monthly status of the college enrollment.
 - education: highest degree earned.
- ▶ **Sample:** 2,489 individuals with non-missing belief, degree, skill, and gender.

Summary Statistics by Gender

here are

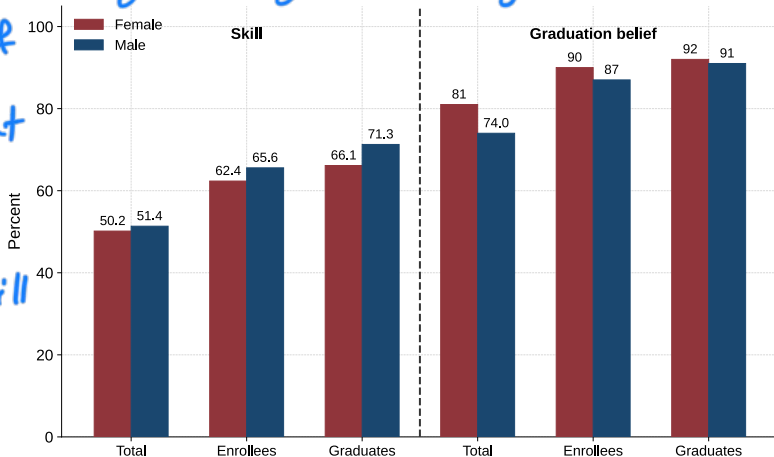


► Males: lower beliefs, lower enrollment, lower graduation, larger belief gap.

Skill and Beliefs Across the Education Stages

First thing that might come to your mind about belief

*diff
is
that
due
to
skill*

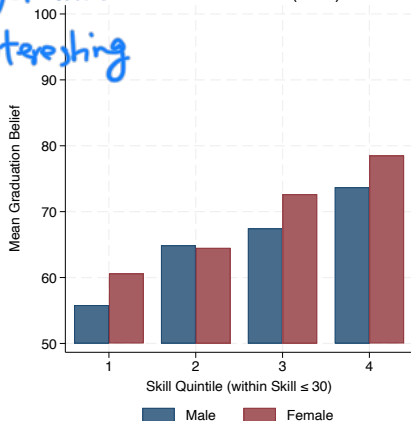


- ▶ College filters men harshly on skill because beliefs failed to filter them at entry.

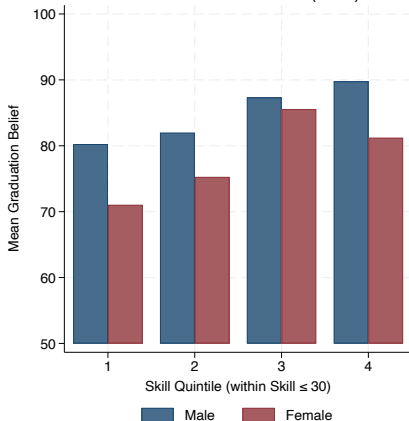
Belief Selection Among Low-Skill Individuals *But when you zoom in things get more interesting*

Beliefs by Skill Among Low-Skill Individuals

Total — Low Skill (≤ 30)



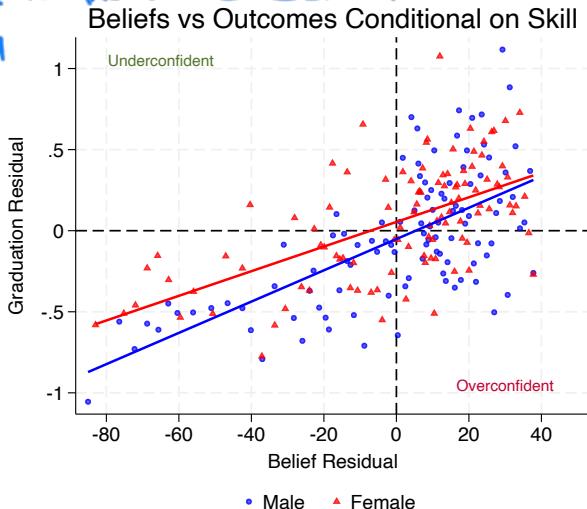
Enrollees — Low Skill (≤ 30)



- Among low-skill enrollees, 81% of males report beliefs above 70% vs. 74% of females.

Beliefs vs Outcomes Conditional on Skill

to give you the big picture here is the whole list of belief & outcome residual



- Male beliefs are **noisier signals** of true ability.

Male Beliefs Are Noisier Signals of Skill

to be more specific

- ▶ **Conditional on skill, male belief variance is 23% higher than female** (SD: 28.9 vs. 26.1; Brown–Forsythe $p < 0.0001$, $N = 2,489$).
- ▶ **The asymmetry intensifies through the pipeline:**

Stage	$\hat{\rho}^m$	$\hat{\rho}^f$	Gap
Total	0.30	0.34	0.04
Enrollees	0.18	0.23	0.05
Graduates	0.12	0.20	0.08
Dropouts	0.10	0.21	0.11

⇒ Male beliefs are **noisier in level** and **progressively less informative** through selection.

Beliefs Drive Enrollment Beyond Skill

lets say beliefs
are naissier for
men
but do
belief
matter
at all

Logit of enrollment on belief and gender, **within each skill tercile**:

	Low Skill (Tercile 1)	Mid Skill (Tercile 2)	High Skill (Tercile 3)
Belief	0.0198*** (0.0033)	0.0298*** (0.0035)	0.0404*** (0.0051)
Female	0.0681 (0.1667)	0.2359 (0.1523)	0.2889 (0.1988)
Constant	-2.5708*** (0.2707)	-2.2984*** (0.3007)	-2.1900*** (0.4339)
Pseudo R^2	0.056	0.112	0.141
N	830	830	829

- ▶ Beliefs are **highly significant** for enrollment at every skill level — not just proxying for ability.
- ▶ The belief coefficient **increases with skill**: beliefs matter more when constraints are less binding.

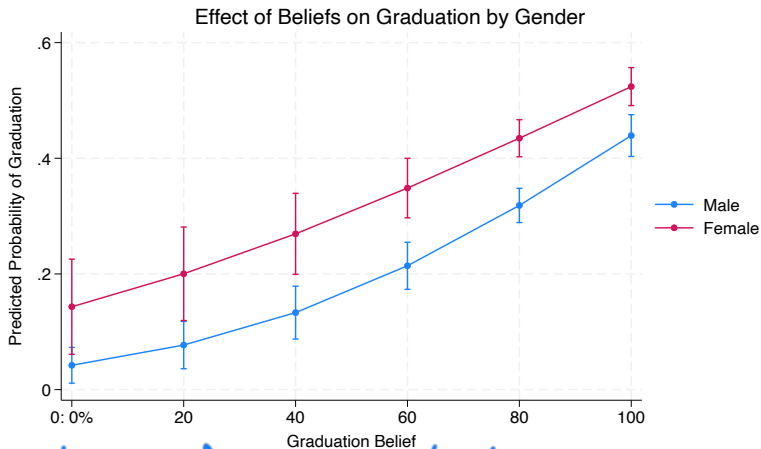
Effect of Beliefs on Enrollment and Graduation by Gender

	Enrollment	College Degree			
	(1)	(2)	(3)	(4)	(5)
Belief	0.0295*** (0.0030)	0.0312*** (0.0027)	0.0393*** (0.0042)	0.0290*** (0.0028)	0.0352*** (0.0043)
Female	1.0633*** (0.3538)	0.2731*** (0.1005)	1.6029*** (0.4895)	0.5497*** (0.1149)	1.5728*** (0.5126)
Belief × Female	-0.0086** (0.0041)		-0.0150*** (0.0054)		-0.0116** (0.0057)
Skill	0.0334*** (0.0021)	0.0355*** (0.0020)	0.0357*** (0.0020)	0.0286*** (0.0022)	0.0288*** (0.0022)
Effort	0.0053 (0.0456)	0.0706 (0.0466)	0.0686 (0.0471)	0.0925* (0.0481)	0.0907* (0.0481)
Risk	0.0900*** (0.0215)	0.0530** (0.0224)	0.0532** (0.0224)	0.0529** (0.0229)	0.0532** (0.0229)
Constant	-5.4290*** (0.6381)	-5.3128*** (0.2630)	-6.0284*** (0.3937)	-5.6919*** (0.7573)	-6.2713*** (0.8067)
Controls	Yes	No	No	Yes	Yes
Pseudo R^2	0.289	0.243	0.245	0.301	0.303
N	2,398	2,489	2,489	2,401	2,401

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Std. errors in parentheses. Controls: marital status, household income, parent BA, pregnancy, number of children.

beliefs seems to have different effect on fem
8 me

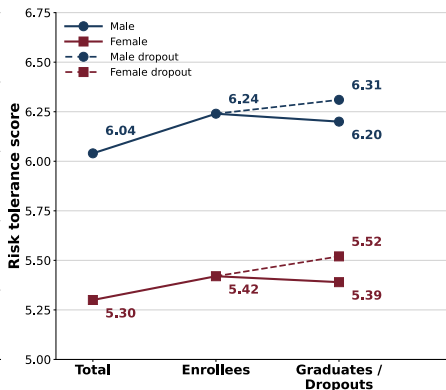
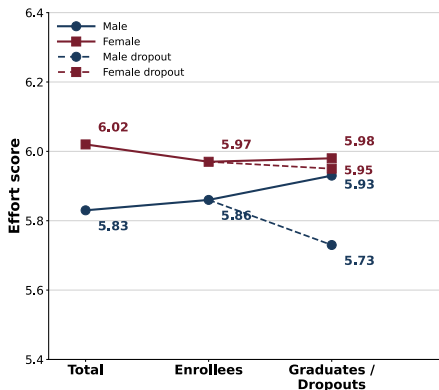
Effect of Beliefs on Graduation by Gender



but why females graduate more no matter what

- Predictive margins from the interaction model with controls.

Effort and Risk Tolerance by Gender and Stage



- ▶ Behavioral channels differ by gender: male dropout is associated with **low effort**.
- ▶ Effort is **independent** of both skill and beliefs for both genders:
 $\text{Corr}(e_i, s_i) \approx 0$ and $\text{Corr}(e_i, \pi_i) \approx 0$.

so far data shows that

- Male beliefs are noisier
- belief matter for edu choice

Model

Environment and Timing

- ▶ **Two generations:** Old parents and young agents with gender $g \in \{m, f\}$ and type (s_i, e_i, π_i^g, b_i) .
- ▶ **Period 0:** Parent transfers b_i . Agent enrolls (pay tuition ϕ) or works.
- ▶ **Period 1:** Graduates earn $w_h \cdot s_i$; non-graduates earn w_l^g .

Outcome	c_0	c_1
Never enroll	$b_i + w_l^g$	w_l^g
Enroll + graduate	$b_i - \phi$	$w_h \cdot s_i$
Enroll + drop out	$b_i - \phi$	w_l^g

Beliefs and Graduation Technology

Subjective belief formation (linear signal with gender-specific noise):

$$\pi_i \mid s_i, g = \mu^g(s_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, (\sigma_\epsilon^g)^2)$$

where $\mu^g(s_i)$ is the gender-specific mean belief given skill and σ_ϵ^g is **belief noise**.

Key assumption: Male beliefs are **noisier signals** of skill:

$$\sigma_\epsilon^m > \sigma_\epsilon^f > 0$$

True graduation probability (common technology, increasing in skill and effort): $p(s_i, e_i)$

Drives enrollment

π_i (subjective belief draw)
Individual does not know $p(\cdot)$

Drives graduation

$p(s_i, e_i)$ (true probability)
Revealed after enrollment

Production: CES Final Good

A competitive firm produces the final good using high-skill and low-skill labor:

$$Y = \left[\alpha L_h^{\frac{\gamma-1}{\gamma}} + (1-\alpha) L_l^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$$

where:

- $L_h = \sum_{i \in \text{grad}} s_i$ (effective high-skill labor, ability-weighted)
- $L_l = N_l$ (low-skill labor: non-graduates + dropouts)
- $\gamma > 0$: elasticity of substitution; $\alpha \in (0, 1)$: share parameter

Endogenous wages from profit maximization:

$$w_h = \alpha \left(\frac{Y}{L_h} \right)^{1/\gamma}, \quad w_l = (1-\alpha) \left(\frac{Y}{L_l} \right)^{1/\gamma}$$

Gender-specific low-skill wages:

$$w_l^m = w_l, \quad w_l^f = (1-\tau) w_l$$

Young Agent's Enrollment Decision

CRRA utility: $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, $\sigma > 0$; lifetime utility $U_i = u(c_0) + \beta u(c_1)$.

Value if not enrolling:

$$U_i^{no} = u(b_i + w_l^g) + \beta u(w_l^g)$$

Expected value if enrolling:

$$EU_i^{enroll} = u(b_i - \phi) + \beta \left[\pi_i u(w_h s_i) + (1 - \pi_i) u(w_l^g) \right]$$

Enrollment rule: enroll if $EU_i^{enroll} > U_i^{no}$:

$$\pi_i > \bar{\pi}^g(s_i, b_i) = \frac{(b_i + w_l^g)^{1-\sigma} - (b_i - \phi)^{1-\sigma}}{\beta \left[(w_h s_i)^{1-\sigma} - (w_l^g)^{1-\sigma} \right]}$$

where $w_l^g = (1 - \tau^g) w_l$.

- ▶ Higher b_i or $s_i \Rightarrow$ **lower cutoff** (richer / more able students need less confidence).
- ▶ Higher $\tau \Rightarrow$ **lower cutoff for females** — worse outside option makes college more attractive.

Parent's Problem

The old parent has income w_p , belief π_p about the child's graduation, and altruism $\lambda \in (0, 1)$. Chooses transfer b_i .

Parent's problem:

$$\max_{b_i \geq 0} u(w_p - b_i) + \lambda V_c^p(b_i; \pi_p)$$

Child's value function

$$V_c^p(b_i; \pi_p) = u(b_i - \phi) + \beta \left[\pi_p u(w_h s_i) + (1 - \pi_p) u(w_l^g) \right]$$

Optimal transfer:

$$b_i^* = \frac{\lambda^{1/\sigma} w_p + \phi}{1 + \lambda^{1/\sigma}}$$

- Higher $\lambda \Rightarrow$ **larger transfer**; higher $\phi \Rightarrow$ **larger transfer** (parent covers more tuition).

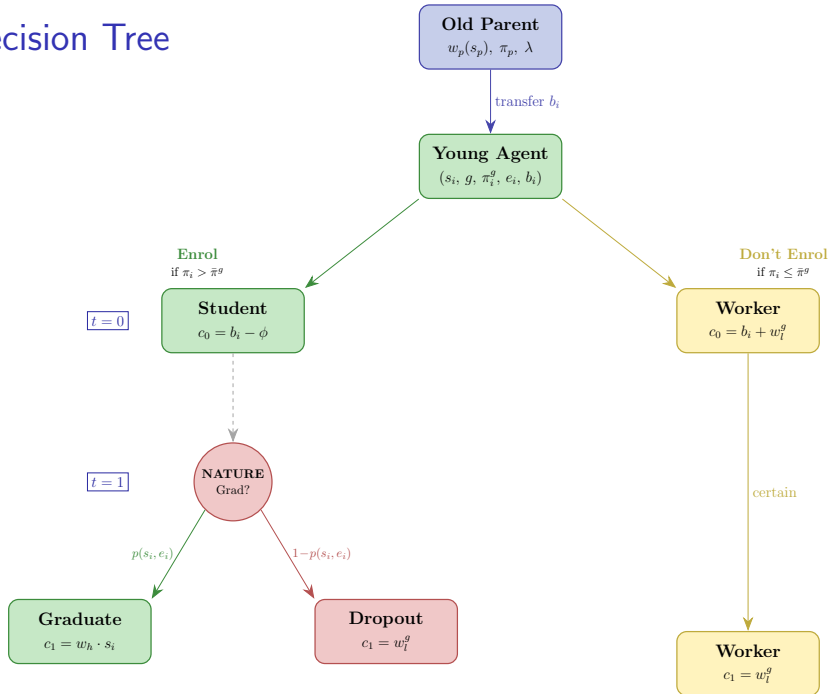
General Equilibrium

Definition. Given the distribution of types (s_i, e_i, π_i^g, g) , A general equilibrium consists of wages (w_h, w_l) , parental transfers $\{b_i^*\}$, enrollment decisions $\{d_i^*\}$ such that:

- (i) Given wages, parents choose optimal transfer b_i^*
- (ii) Given (b_i^*, w_h, w_l) , agents enroll ($d_i^* = 1$) iff $\pi_i > \bar{\pi}^g(s_i, b_i^*)$.
- (iii) Wages solve the firm's problem.
- (iv) Labor markets clear:

$$L_h = \sum_{i \in \text{grad}} s_i, \quad L_l = |\{i : \text{non-grad or dropout}\}|.$$

Decision Tree



How the Model Generates the Key Facts

Enrollment and Dropout

Fact 1: Female over-representation in enrollment ($E^f > E^m$).

- ▶ Higher female beliefs and the wage barrier τ both **lower** $\bar{\pi}^f \Rightarrow$ more females enroll.

Fact 2: Higher male dropout rates ($D^m > D^f$).

- ▶ Lower beliefs $\Rightarrow$ higher skill cutoff $\Rightarrow$ pushes $E[s_i \mid \text{enrolled}, m]$ **up**.
- ▶ Noisier male beliefs ($\sigma_\epsilon^m > \sigma_\epsilon^f$) let a **low-skill tail of males** through the enrollment cutoff.
- ▶ Combined with **lower male effort**, and since skill and effort are substitutes, these mismatched enrollees fail to graduate.

Takeaway and Future Work

Takeaway

- ▶ The gender gap in college completion is not just about **who enrolls** — it is also about *why* they enroll.
- ▶ Both genders **overestimate** graduation chances, but males do so by more, especially **low-skill males**.
- ▶ Beliefs independently drive enrollment at every skill level.
- ▶ The same skill maps to **different belief distributions** by gender: males draw noisier beliefs.
- ▶ Asymmetric **belief noise** σ_{ϵ}^g generates a **selection effect**: noisy male beliefs let low-skill males into college, where the reality filter removes them.

Future Work

► Estimate the model using NLSY97 data

- Match observed enrollment, graduation, and dropout patterns.

► Quantifying misbelief

- Measure how much misbelief contributes to gender gaps in enrollment and dropout.

► Counterfactual experiments in GE

- Belief correction, tuition subsidies (ϕ), barrier reduction (τ).
- Quantify wage and welfare effects of eliminating misbeliefs.

► Richer dynamics

- Savings, borrowing, and endogenous belief updating.
- Intergenerational persistence of misbeliefs through parental transfers.(OLG)
- Gender-specific risk aversion.

Thank You

Questions?